Department of Artificial Intelligence & Data Science

DriveAlert AI: A Real-Time Driver Drowsiness Detection System Using Facial Landmarks and Deep Learning

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CERTIFICATE

This is to certify that **Hema S S, Aishwarya K N and Divyadharshini K** from **Third year AI&DS** have successfully completed their work titled **"DriveAlert AI: A Real-Time Driver Drowsiness Detection System Using Facial Landmarks and Deep Learning"** at SRM TRP Engineering College, Trichy in the partial fulfilment of the Bachelor of Technology in Artificial Intelligence & Data Science.

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ABSTRACT

Driver fatigue is one of the major hidden causes of road accidents. A tired driver reacts slowly, loses lane discipline and may fall asleep for a few seconds, which at highway speed is enough to cause a fatal crash. Unlike drink driving or speeding, drowsiness is difficult to measure and is rarely reported, so there is a strong need for a system that can monitor the driver continuously and warn them before an accident occurs.

This project presents DriveAlert AI, a real-time driver drowsiness detection system that analyses video from a dashboard-mounted camera. The system detects the driver's face using OpenCV and extracts 68 facial landmarks. From the landmarks it computes the Eye Aspect Ratio (EAR) to detect eye closure, the Mouth Aspect Ratio (MAR) to detect yawning, and head-pose angles to detect nodding. A Convolutional Neural Network (CNN) classifies each eye crop as open or closed, which makes the system robust to spectacles and varied lighting. These cues are combined over a sliding time window into drowsiness indicators such as PERCLOS (percentage of eyelid closure), blink duration, yawn frequency and head-nod count. A rule-based fusion stage then classifies the driver's state as Alert, Fatigued or Drowsy, and an audio-visual alarm is raised when a dangerous level is reached.

The CNN is trained on open-eye and closed-eye datasets such as the MRL Eye Dataset and evaluated using accuracy, precision, recall, F1-score and confusion matrices, while the complete pipeline is tested on video datasets such as NTHU-DDD and YawDD. The system runs in real time on an ordinary laptop or an embedded board, requires no wearable sensor, and stores no video, which preserves driver privacy. The project demonstrates that a vision-based approach is a low-cost and practical way to reduce fatigue-related accidents, while factors such as night-time lighting, sunglasses, camera placement and driver variability must be handled carefully.

INTRODUCTION

Introduction: DriveAlert AI – Driver Drowsiness Detection

Road transport is central to modern life, yet road accidents remain a leading cause of death and injury worldwide. Studies by road-safety agencies consistently identify driver fatigue as a significant contributing factor, especially on long highway journeys, night-time driving and in commercial transport such as trucks, buses and taxis. Fatigue slows reaction time, reduces attention and impairs judgement in a manner comparable to alcohol. The most dangerous consequence is a micro-sleep, a brief and involuntary loss of awareness lasting a few seconds, during which a vehicle can travel a long distance without any control.

Existing approaches to detecting drowsiness fall into three groups. Vehicle-based methods monitor steering behaviour or lane deviation, but they react only after driving performance has already degraded and depend on road and vehicle conditions. Physiological methods measure signals such as EEG, ECG or heart-rate variability and are accurate, but they need electrodes or wearables that are intrusive and uncomfortable. Behavioural, vision-based methods observe the driver's face and are non-intrusive, inexpensive and able to detect fatigue early, which is why they are chosen for this project.

The visible signs of drowsiness are well documented: eyelids stay closed longer and blinks become slow, the driver yawns frequently, and the head droops or nods forward. A single frame cannot reliably capture these signs, because a normal blink looks identical to the start of a micro-sleep. Drowsiness is therefore a property of behaviour over a period of time and must be measured across a window of frames.

DriveAlert AI addresses this by combining geometric facial analysis with deep learning. Facial landmarks provide precise measurements of the eyes, mouth and head orientation, a CNN gives a robust open/closed decision for each eye, and a temporal fusion module turns these per-frame observations into a drowsiness level. The problem is divided into four tasks: face and landmark detection, eye-state and yawn analysis, drowsiness-level estimation, and driver alerting with event logging.

The quality of the training data strongly affects performance. Public datasets such as the MRL Eye Dataset, which contains infrared eye images in many conditions, and NTHU-DDD and YawDD, which contain videos of drivers in realistic situations including glasses, sunglasses, yawning and night-time, provide a wide variety of faces and lighting. Preprocessing and data augmentation are applied to make the model robust to in-vehicle conditions.

Several challenges must be addressed: changing illumination from sunlight and street lights, reflections on spectacles, occlusion by sunglasses or a hand, different camera angles, and individual differences in eye shape and blinking habits. Driver privacy is also important, so the system processes video locally and stores only event logs rather than recordings. The following chapters describe the requirements, methodology and implementation of the system.

SYSTEM REQUIREMENTS

Category	Requirement	Description
Operating System	Windows / Linux / macOS	Any OS supporting Python and deep learning frameworks. Raspberry Pi OS can be used for an in-vehicle prototype.
Processor (CPU)	Intel i5 / AMD Ryzen 5 or higher	Needed for real-time frame processing and landmark detection.
GPU (Optional)	NVIDIA GTX 1650 or higher	Speeds up CNN training; inference runs in real time on CPU.
RAM	Minimum 8 GB (16 GB recommended)	Handles datasets, batch training and video buffers.
Storage	Minimum 20 GB free space	Stores datasets, trained models and project files.
Camera	HD webcam / infrared (IR) USB camera, 720p, 30 FPS	Captures the driver's face; IR gives reliable night-time operation.
Alert Hardware	Speaker / buzzer	Produces an audible alarm when drowsiness is detected.
Programming Language	Python 3.x	Simple, with extensive library support.
Deep Learning Framework	TensorFlow / Keras	Used to build, train and evaluate the eye-state CNN.
Computer Vision Library	OpenCV, dlib / MediaPipe	Used for face detection, 68-point landmark extraction and head-pose estimation.
Data Handling Libraries	NumPy, Pandas, SciPy	Used for data processing and distance calculations for EAR and MAR.
Visualisation & Dashboard	Matplotlib, Plotly, Streamlit	Used for training curves and the monitoring dashboard.
Audio Library	pygame / playsound	Plays the alarm sound.
Development Environment	Visual Studio Code / Jupyter Notebook	Used for writing, testing and debugging.
Dataset	MRL Eye Dataset, NTHU-DDD, YawDD	Labelled eye images and driver videos for training and testing.

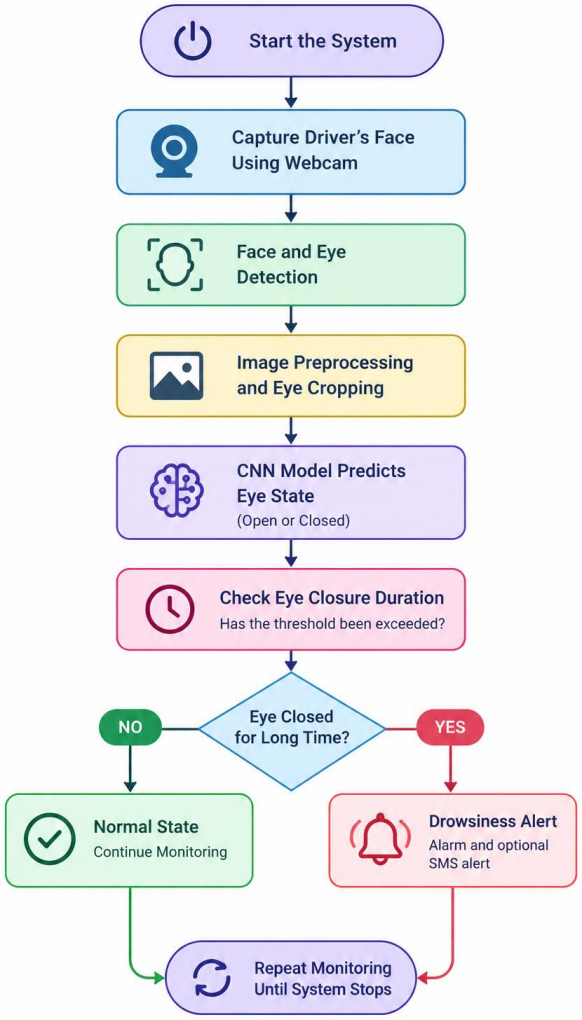
METHODOLOGY

Methodology: DriveAlert AI

The methodology follows a structured pipeline: video capture, face detection and landmark extraction, feature computation, eye-state classification, temporal analysis, drowsiness classification, alerting and evaluation. Geometric features from facial landmarks are combined with a CNN-based eye-state classifier, and the evidence is accumulated over time so that a decision is made from sustained behaviour rather than a single frame.

1. Overall System Workflow

The system captures live video from a camera facing the driver. Each frame is converted to grayscale, the face is detected, and 68 facial landmarks are located. From the landmarks, the eye and mouth regions and head pose are measured. The eye crops are passed to the CNN, and the results are accumulated in a sliding window. The fusion module decides whether the driver is Alert, Fatigued or Drowsy, and triggers the alarm and logs the event when required.



2. Data Collection and Dataset Selection

Three kinds of data are used. The MRL Eye Dataset provides a large number of cropped eye images labelled open or closed, captured with different sensors, in different lighting and with or without glasses, and is used to train the eye-state CNN. NTHU-DDD (NTHU Driver Drowsiness Detection) provides videos of drivers simulating normal driving, slow blinking, nodding and yawning, with and without glasses, in day and night conditions, and is used to evaluate the complete pipeline. YawDD provides videos of drivers yawning, talking and driving normally, and is used to tune the yawn detector. Additional images can be recorded from volunteers in the college laboratory to test the system under local conditions.

Key dataset characteristics:

- Large number of labelled eye images and driver videos
- Variation in lighting, glasses, head pose and camera position
- Examples of normal blinking, long eye closure, yawning and nodding
- Day and night (infrared) conditions for better generalisation

A balanced dataset is essential to avoid bias, so the number of open-eye and closed-eye samples is equalised before training.

3. Data Preprocessing

Preprocessing makes every input consistent before it is passed to the models.

- **Face Detection:** OpenCV detects the face region in every frame.
- **Grayscale Conversion and Resizing:** Frames are converted to grayscale and eye crops are resized to 64×64 pixels.
- **Normalisation:** Pixel values are scaled between 0 and 1.
- **Histogram Equalisation:** CLAHE improves contrast in dark or unevenly lit frames.
- **Data Augmentation:** Rotation, flipping, brightness change, zoom and blur are applied to the eye images to make the model robust.
- **Dataset Split:** Data is divided into training, validation and testing sets, keeping images of the same person in only one set to prevent leakage.

4. Face Detection and Facial Landmark Extraction

The face is located with a Haar cascade or HOG-based detector, and a pretrained shape predictor then marks 68 landmark points around the eyebrows, eyes, nose, mouth and jawline. The six points around each eye give the eye region used for the Eye Aspect Ratio and for cropping the eye image, and the points around the lips give the mouth region used for yawn detection. If no face is found for several consecutive frames, the system warns that the driver is not visible, since an absent face can itself indicate a dangerous situation.

5. Drowsiness Feature Extraction

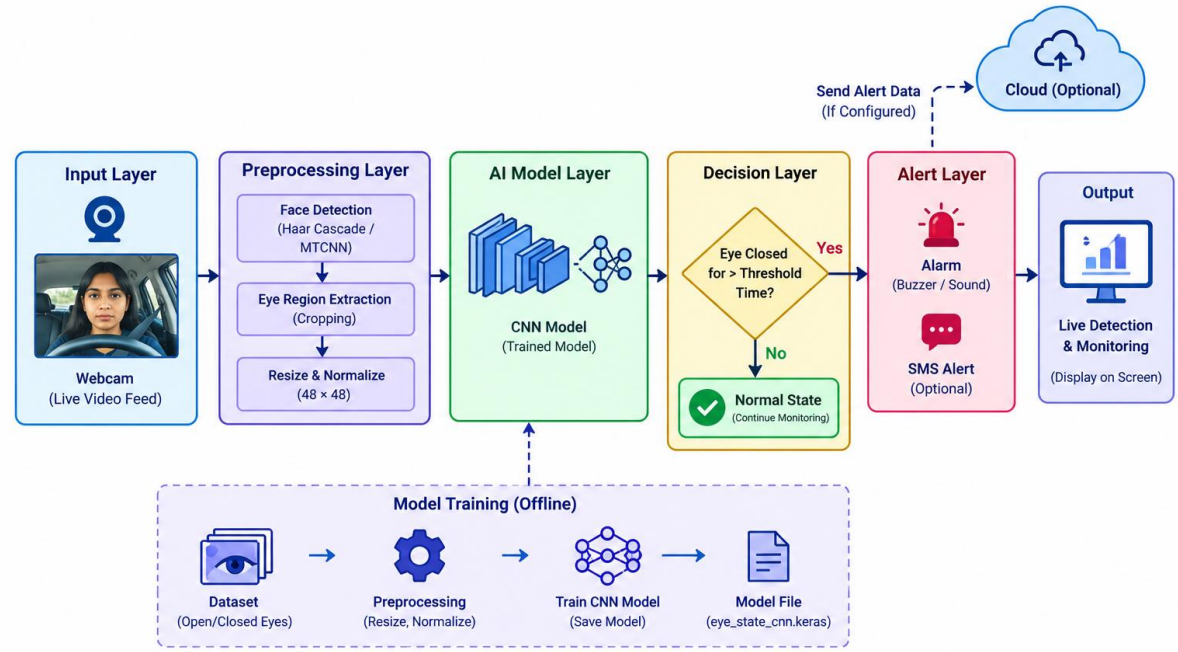
The following features are computed for each frame.

- **Eye Aspect Ratio (EAR):** The ratio of the vertical eye distances to the horizontal eye width. EAR stays roughly constant while the eye is open and falls towards zero when it closes. $EAR = (||p2 - p6|| + ||p3 - p5||) / (2 \times ||p1 - p4||)$.
- **Mouth Aspect Ratio (MAR):** The ratio of the vertical mouth opening to the mouth width. A high MAR sustained for more than a second indicates a yawn.
- **Head Pose:** Pitch, yaw and roll angles are estimated from selected landmarks using solvePnP. A sustained downward pitch indicates head nodding.
- **Blink Features:** Blink rate and blink duration. Long, slow blinks are a typical early sign of fatigue.
- **PERCLOS:** The percentage of time within a window (for example 60 seconds) during which the eyes are at least 80% closed, a widely used and validated indicator of drowsiness.

6. CNN-Based Eye State Classification

Geometric thresholds such as EAR can fail with glasses, different eye shapes or poor landmark accuracy. To improve robustness, a small CNN classifies each cropped eye as open or closed. The network contains three convolution and max-pooling blocks with 32, 64 and 128 filters, followed by a flatten layer, a dense layer of 128 units with dropout, and a final sigmoid

output. The CNN probability is combined with the EAR value, so that the eye is declared closed only when both cues agree, which reduces false alarms.



[Figure 3: CNN architecture for eye-state classification – diagram to be inserted]

Model components:

- **Convolutional Layers:** Extract edges, eyelid contours and texture
- **Pooling Layers:** Reduce dimensionality and computation
- **Dropout Layer:** Reduces overfitting
- **Fully Connected Layers:** Produce the open/closed prediction

7. Model Training

The CNN is trained on the preprocessed eye images, with the validation set used to monitor overfitting.

Key training components:

- **Loss Function:** Binary cross-entropy
- **Optimizer:** Adam optimizer with a small learning rate
- **Batch Size and Epochs:** Tuned according to validation performance
- **Regularisation:** Dropout and early stopping
- **Callbacks:** Model checkpointing to save the best weights and learning-rate reduction on plateau

Training uses backpropagation to minimise the classification error between predicted and true eye states.

8. Drowsiness Decision Logic

The per-frame outputs are accumulated over a sliding time window and the driver's state is decided by rules combining several indicators.

- **Alert:** PERCLOS is low, blinks are short and no yawns or nods are detected.
- **Fatigued (Warning):** PERCLOS rises above a first threshold, or repeated yawns or slow blinks occur. A soft warning is shown.
- **Drowsy (Danger):** Eyes remain closed for more than about two seconds, PERCLOS exceeds the danger threshold, or a sustained head nod is detected. A loud alarm is sounded.

The thresholds are calibrated for each driver during the first few seconds of driving by recording their normal open-eye EAR, which accounts for differences in eye size.

9. Model Evaluation

The eye-state CNN is evaluated on the unseen test set using accuracy, precision, recall and F1-score. The complete drowsiness pipeline is evaluated on driver videos by comparing detected states with annotated labels. For safety applications, recall for the drowsy class is especially important because a missed detection is more costly than a false alarm, while the false alarm rate is also monitored so that the driver does not ignore the system. Confusion matrices and training and validation curves are used to analyse errors, for example blinks confused with micro-sleeps or talking confused with yawning. The processing speed in frames per second is also measured to confirm real-time operation.

10. Alert System and Monitoring Dashboard

When a warning or danger state is reached, the system displays a message on screen and plays an alarm through the speaker. Each event is logged with its time and type, and a dashboard summarises the live EAR and MAR values, PERCLOS, blink rate, yawn count, current driver state and a timeline of alerts during the trip. Trip reports can help fleet operators identify risky time periods and plan rest breaks.

11. Prediction and Deployment

Once trained and evaluated, the system runs in real time on a live camera feed. The steps are:

- Capture of video frames from the driver-facing camera
- Face detection, landmark extraction and preprocessing
- EAR, MAR and head-pose computation and CNN eye-state prediction
- Window-based decision and alert generation

The system can be deployed on a laptop or on a low-cost embedded board such as a Raspberry Pi with an infrared camera, and all processing is done on the device, so no video needs to leave the vehicle.

12. Challenges and Optimisation

- Variation in lighting, including sunlight, shadows and headlights at night
- Spectacle reflections, sunglasses and partial occlusion of the face
- Differences in eye shape, blink habits and camera position between drivers
- Distinguishing normal blinks, talking and looking at mirrors from real drowsiness

- Driver privacy and consent

To overcome these challenges, techniques such as data augmentation, infrared imaging, per-driver calibration, temporal smoothing, threshold tuning and lightweight model design can be applied, and only event logs, not raw video, are stored.

13. Summary of Methodology

The methodology covers a complete pipeline from video capture and preprocessing to landmark extraction, feature computation, CNN-based eye-state classification, temporal decision-making, alerting and evaluation. By combining geometric measures with a learned eye-state classifier and judging behaviour over a time window, the system detects early signs of fatigue and warns the driver before a dangerous situation develops.

IMPLEMENTATION

Implementation of DriveAlert AI

DriveAlert AI is implemented in Python. TensorFlow/Keras is used for the eye-state CNN, OpenCV and dlib (or MediaPipe) for video processing and facial landmarks, SciPy for distance calculations, and Streamlit with Matplotlib and Plotly for the dashboard. The project is organised into small modules, each responsible for one stage of the pipeline.

Project modules:

- **face_utils.py:** Reads video frames, detects the face, extracts the 68 landmarks and crops the eye regions.
- **features.py:** Computes EAR, MAR, head-pose angles, blink duration and PERCLOS over a sliding window.
- **eye_model.py:** Defines the CNN used to classify each eye as open or closed.
- **train.py:** Loads the eye dataset, applies augmentation, trains the CNN and saves the best weights.
- **decision.py:** Fuses the indicators and assigns the Alert, Fatigued or Drowsy state.
- **alert.py:** Plays the alarm and records events in a log file.
- **app.py:** Runs the live pipeline and shows the results on the dashboard.

Execution steps:

- The driver-facing camera is opened and frames are read continuously.
- The face and landmarks are detected, and the system is calibrated on the driver's normal eye opening.
- EAR, MAR and head pose are computed, and eye crops are classified by the CNN.
- Indicators are accumulated over the window and the driver's state is decided.
- If the state is Fatigued or Drowsy, a warning or alarm is raised and the event is logged.

The source code of the model is shown in Annexure A, the monitoring dashboard is shown in Annexure B, and the drowsiness detection output is shown in Annexure C.

CONCLUSION

The implementation of DriveAlert AI demonstrates that a low-cost, camera-based system can monitor a driver continuously and detect the early signs of drowsiness in real time. By combining facial-landmark measures such as EAR, MAR and head pose with a CNN that classifies eye state, the system avoids relying on a single cue and is more robust to glasses, lighting changes and individual differences. Judging behaviour over a time window through indicators such as PERCLOS allows it to distinguish a harmless blink from a dangerous eye closure. A structured methodology was followed, from dataset selection and preprocessing to model training, decision logic, alerting and evaluation, and the alert dashboard gives a clear record of each trip.

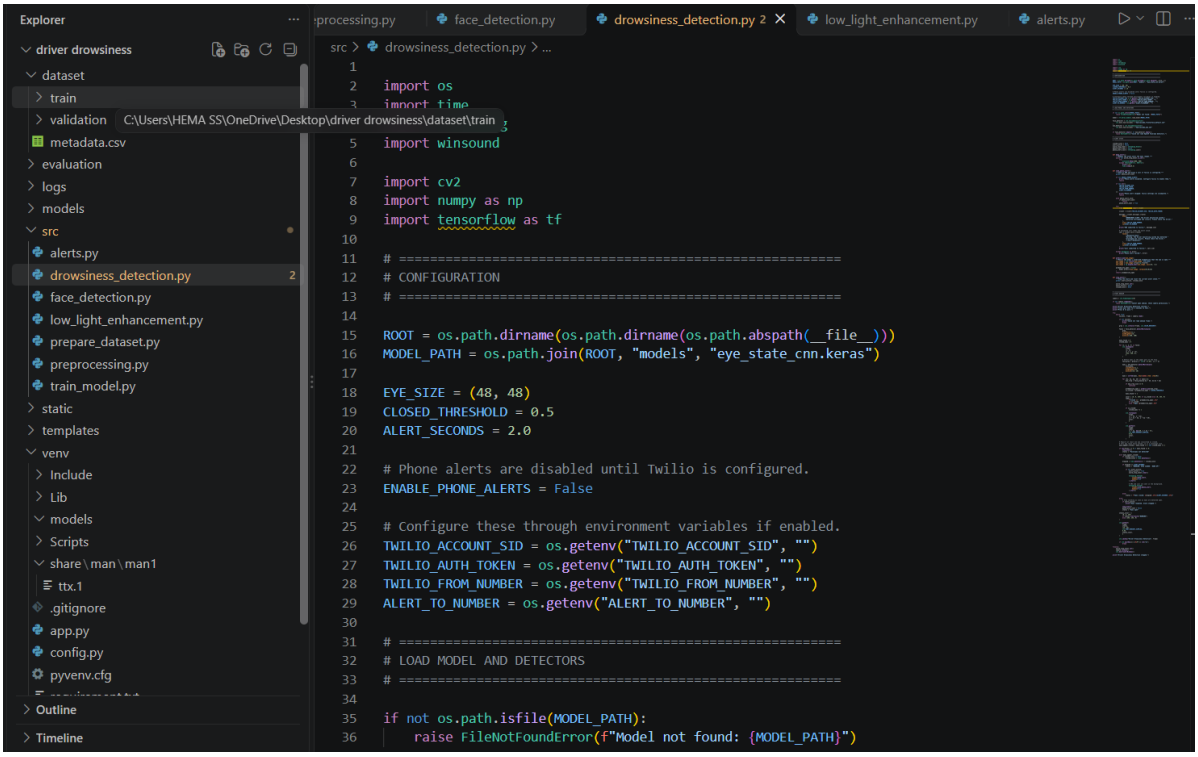
In the future, the system can be extended with infrared cameras for reliable night-time operation, gaze and distraction detection, a lightweight temporal model such as an LSTM for learning drowsiness patterns directly from sequences, integration with vehicle signals such as steering and lane data, and mobile or cloud connectivity so that fleet managers can receive alerts. Throughout, the system should be used responsibly, with the driver's awareness and consent, as a safety aid that supports the driver and does not replace the need for adequate rest.

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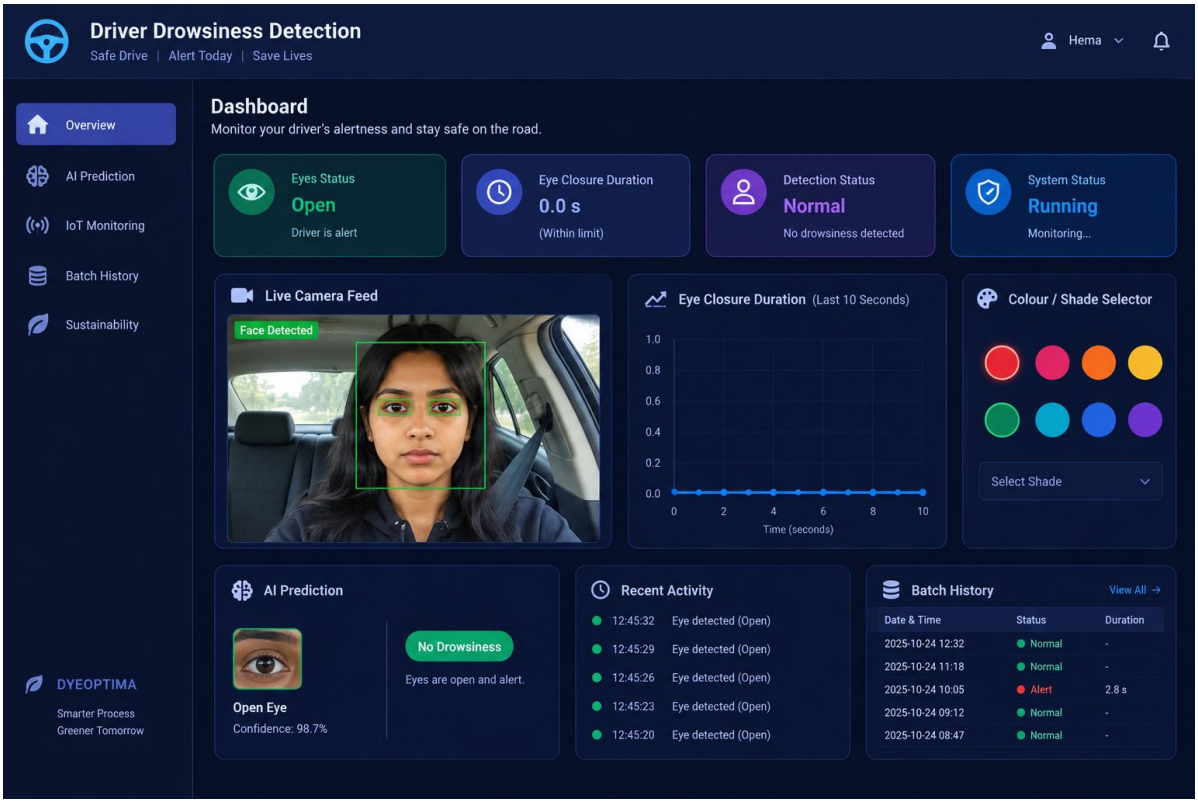
ANNEXURE A – SOURCE CODE

The screenshot below shows the implementation of the eye-state CNN and the EAR calculation in Visual Studio Code.



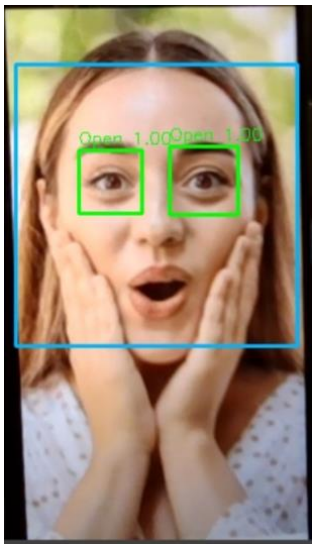
ANNEXURE B – DRIVER MONITORING DASHBOARD

The dashboard summarises a driving session: live EAR and MAR values, PERCLOS, blink rate, yawn count, the current driver state, and a timeline of warnings and alarms.

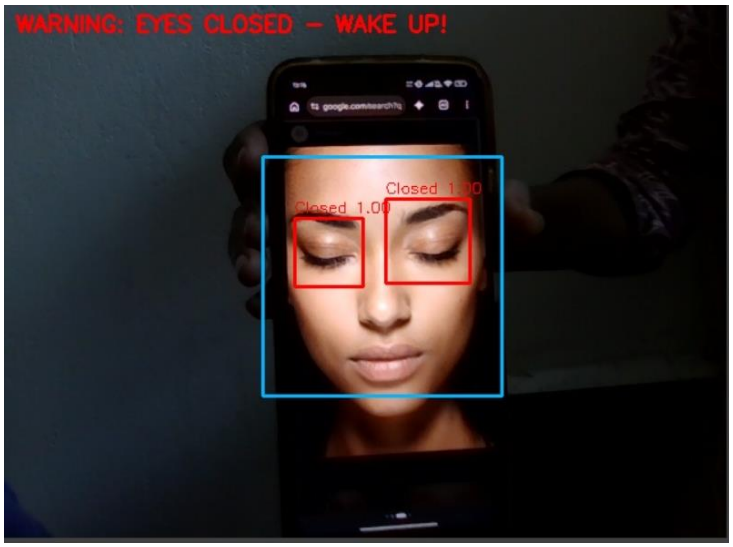


ANNEXURE C – DROWSINESS DETECTION OUTPUT

The figures show the detection stage running on a webcam in Visual Studio Code. The face is enclosed in a bounding box, the eyes and mouth are outlined with landmarks, and the driver's state (Alert / Drowsy) is displayed with the EAR value.



(a) Alert state: eyes open



(b) Drowsy state: eyes closed, warning displayed